

AI-supported Nishijin-ori: connecting a text-to-image model to traditional Nishijin-ori textile production

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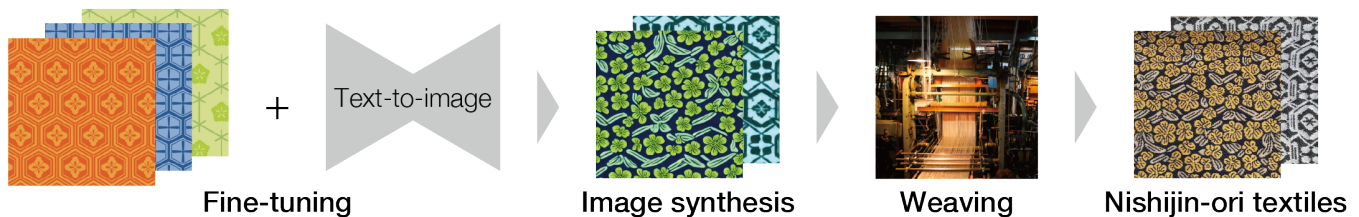


Figure 1: Overview of AI-supported Nishijin-ori. We first generated patterns with a fine-tuned text-to-image model and then produced traditional woven Japanese textiles, Nishijin-ori textiles.

CCS CONCEPTS

• Applied computing → Arts and humanities.

KEYWORDS

Text-to-image model, Nishijin-ori, Geometric patterns, Textiles

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1 INTRODUCTION

Text-to-image models based on diffusion models have recently made significant progress. Such models enable us to accomplish various tasks, including the generation of high-quality images from given text prompts, image inpainting, image outpainting, and super-resolution. Indeed, in some industries, such as art and advertising, generative models have begun to be effectively utilized. However, generative models achieving human-comparable performance are

giving rise to new issues, such as the displacement of jobs in relevant fields. People are now exploring ways to promote harmonious coexistence and collaboration with these generative models.

In this study, we integrated a state-of-the-art text-to-image model and Nishijin-ori textile production, a traditional industry in Japan. As shown in Figure 1, we first generated pattern images using a text-to-image model fine-tuned with existing Yūsoku monyō pattern images. Yūsoku monyō is a pattern that mainly developed during the Heian period (794–1185) and is currently used in the designs of Kimonos, architecture, and pottery. We then created Nishijin-ori textile samples and a Kimono using a weaving machine. Nishijin-ori is a traditional woven textile developed in the Nishijin district of Kyoto. It was initially created with Jacquard looms, which are not only culturally important but also historically significant in that they have led to the development of computers. This study presents an example of connecting generative models to traditional cultures.

2 METHODS

Large datasets of Yūsoku monyō pattern images do not exist, and public pre-trained text-to-image models are insufficient for generating such pattern images. To generate pattern images of Nishijin-ori textiles, we first fine-tuned the publicly available pre-trained text-to-image model with existing patterns. As existing patterns, we used Yūsoku monyō pattern images included in a book [Hachijou 2009]. This study focused on one of the representative Yūsoku monyō patterns, Kikkō pattern, which resembles the hexagonal patterns of turtle shells, as shown in Figure 2.

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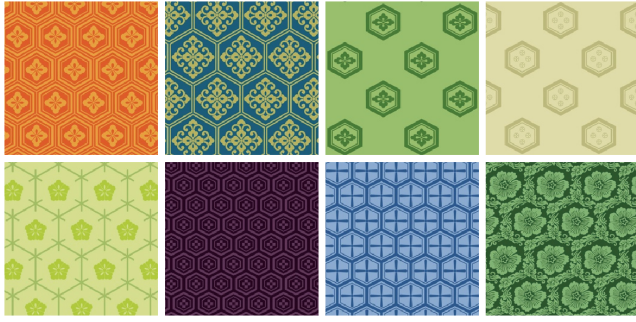


Figure 2: Kikkō pattern images used for fine-tuning a text-to-image model in this study. Images are from [Hachijou 2009]

We fine-tuned a text-to-image model following DreamBooth [Ruiz et al. 2023], which enables fine-tuning of text-to-image models with a few images. We used the eight Kikkō pattern images in Figure 2 and a text prompt with a unique identifier ("a [V] pattern"). As a base text-to-image model, we used Stable diffusion v1-4¹ [Rombach et al. 2022]. To generate tileable patterns, we used circular padding in the convolutional layers of the model.

We generated Kikkō pattern images with the fine-tuned text-to-image model. The generated pattern images were tiled to create larger images and then manually converted into a bitmap format. Based on these bitmap images, we produced Nishijin-ori textile samples using a weaving machine at Fukuoka Weaving Co., Ltd. After adjusting the color scheme, we produced a Kimono.

3 RESULTS

Figure 3 shows example images generated using the fine-tuned text-to-image model. Generating these patterns using public models is challenging because such types of images are not often included in the original training datasets. However, we fine-tuned a text-to-image model with Kikkō images; consequently, the generated patterns exhibited characteristics of the hexagonal structure of the Kikkō pattern. Specifically, Figure 3 (b) shows a regular arrangement of hexagons, whereas Figures 3 (a) and (c) show partial hexagonal structures with added randomness. Figure 4 (a) shows a photograph of Nishijin-ori textile samples produced using a weaving machine. These textiles were produced based on the two selected generated patterns, (a) and (b) in Figure 3. Additionally, Figure 4 (b) shows a photograph of Nishijin-ori Kimono based on design (a) in Figure 3.

4 DISCUSSION AND CONCLUSIONS

Our study materialized the generated images, revealing the differences between the generated images and the corresponding materialized textiles, particularly in terms of color and size. We are considering further analysis and sensing with cameras to construct a generating loop that includes the materials. The construction of a material-in-the-loop may enable more desirable materialization, which might encompass not only patterns and colors, but also the tactile properties of the material.

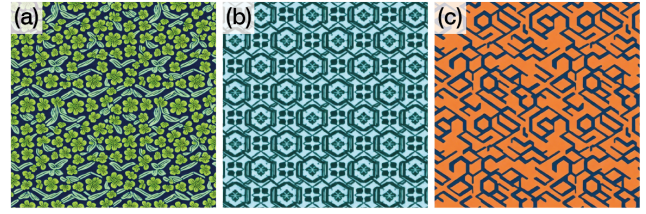


Figure 3: Example images generated using the fine-tuned text-to-image model with the following prompts: (a) "a green [V] pattern with small yellow flowers," (b) "a crisp geometric [V] pattern," and (c) "a pastel orange geometric [V] pattern."

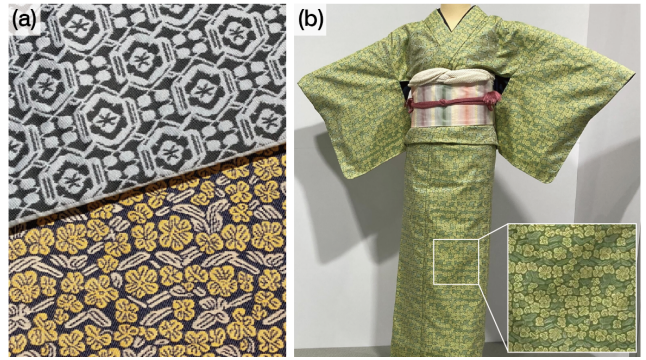


Figure 4: (a) Nishijin-ori textile samples of generated Kikkō pattern images, (a) and (b) in Figure 3. (b) Nishijin-ori Kimono based on design (a) in Figure 3.

Many manufacturing processes are involved in the production of Nishijin-ori textiles, including design creation, pattern paper creation, yarn dyeing, and weaving. As a first step, this study presents a workflow for producing Nishijin-ori textiles based on designs generated using a text-to-image model. Text-to-image model-based pattern generation is beneficial to designers. For example, in Figure 3 (a), the hexagonal part consisting of leaves in response to the text "small yellow flowers" is creative, indicating that AI can support idea creation. In the future, we will also explore ways to fuse traditional craftsmanship with state-of-the-art technologies in other manufacturing processes for Nishijin-ori textile production.

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¹<https://huggingface.co/CompVis/stable-diffusion-v-1-4-original>